CHAPTER 9

Net Present Value (NPV) & Capital Budgeting

Discounted Cash Flow Sums, Hurdle Rate Sensitivity, Profile Convexity, and Value Maximization Criteria

---

9.1 Definition & The Discounted Cash Flow Criterion

Net Present Value (NPV) is the gold standard decision-support metric in capital allocation and financial engineering. It evaluates an investment opportunity by discounting all future periodic cash inflows and outflows back to time zero at an appropriate risk-adjusted cost of capital (hurdle rate), then subtracting the initial capital expenditure.

Definition 9.1 (The Net Present Value Equation)

Let $\{C_0, C_1, ..., C_T\}$ represent a finite series of periodic cash flows occurring at integer times $t \in \{0, 1, ..., T\}$, where initial outlay $C_0 < 0$. Let $r \geq 0$ denote the hurdle discount rate.
The **Net Present Value** of the project is defined by:

$$\text{NPV}(r) = \sum_{t=0}^{T} \left[C_t / (1 + r)^t \right] = C_0 + \sum_{t=1}^{T} \left[C_t / (1 + r)^t \right]$$

Decision Criteria: $NPV(r) > 0 \Rightarrow$ Accept (Value Creation) | $NPV(r) < 0 \Rightarrow$ Reject (Value Destruction)

9.2 Calculus of the NPV Profile: Slope & Convexity

Derivative Analysis: Viewing NPV as a continuous real function of the discount rate $r \in (-1, \infty)$:

- **First Derivative (Rate Sensitivity):**
  $d(\text{NPV})/dr = -\sum_{t=1}^{T} \left[t \cdot C_t / (1 + r)^{t+1} \right]$.
  For conventional projects (where all subsequent cash flows $C_t > 0$), $d(\text{NPV})/dr < 0$ strictly, confirming NPV is a monotonically decreasing function of rate r.
- **Second Derivative (Convexity):**
  $d^2(\text{NPV})/dr^2 = \sum_{t=1}^{T} \left[t(t+1) \cdot C_t / (1 + r)^{t+2} \right] > 0$.
  Strict positive second derivative proves the NPV profile is **strictly convex** across all positive rates.

Theorem 9.1 (NPV Reinvestment Rate Principle)

The mathematical formulation of NPV implicitly assumes that all intermediate positive cash inflows $C_1, C_2, ..., C_{T-1}$ are reinvested across time at the exact hurdle rate r, not at the project's internal yield rate.

$$\boldsymbol{FV_{\text{terminal}} = \text{NPV}(r) \cdot (1 + r)^T - C_0(1 + r)^T}$$

Significance: This reinvestment assumption makes NPV economically superior to naive yield metrics (such as IRR) when ranking mutually exclusive investment proposals.

9.3 Step-by-Step Worked Examples

Below are two formal case studies solved by hand: evaluating an industrial asset capital acquisition, and resolving the Fisher intersection rate between competing projects.

Example 9.1: Industrial Infrastructure Asset Evaluation

Problem Statement: An industrial firm evaluates acquiring automated manufacturing equipment. The initial expenditure is C_0 = –**INR 1,000,000**. The asset generates five annual cash inflows: C_1 = **INR 250,000**, C_2 = **INR 350,000**, C_3 = **INR 350,000**, C_4 = **INR 300,000**, and C_5 = **INR 200,000**. The firm's cost of capital is R = **10.00% per annum**. Calculate the project NPV and state the investment verdict.

Step-by-Step Analytical Solution:

Step 1 (Discount Inflows at r = 0.10):

- Year 1: $PV_1 = 250{,}000 / (1.10)^1 = 250{,}000 / 1.10000 =$ **INR 227,272.73**
- Year 2: $PV_2 = 350{,}000 / (1.10)^2 = 350{,}000 / 1.21000 =$ **INR 289,256.20**
- Year 3: $PV_3 = 350{,}000 / (1.10)^3 = 350{,}000 / 1.33100 =$ **INR 262,960.18**
- Year 4: $PV_4 = 300{,}000 / (1.10)^4 = 300{,}000 / 1.46410 =$ **INR 204,904.04**
- Year 5: $PV_5 = 200{,}000 / (1.10)^5 = 200{,}000 / 1.61051 =$ **INR 124,184.26**

Step 2 (Sum Inflows & Subtract Outlay):

- Total PV of Inflows: 227,272.73 + 289,256.20 + 262,960.18 + 204,904.04 + 124,184.26 = **INR 1,108,577.41**.
- Net Present Value: NPV = –1,000,000 + 1,108,577.41 = **+INR 108,577.41**.

Step 3 (Financial Decision & Economic Interpretation):

- Verdict: **ACCEPT THE INVESTMENT**.
- *Economic Meaning:* The project recovers all initial capital, covers the full 10% annual cost of capital, and generates an incremental **INR 108,577.41** in pure economic profit today for the enterprise.

Example 9.2: Mutually Exclusive Projects & The Fisher Intersection

Problem Statement: Two competing investments require identical initial outlays of $C_0 = -$**INR 100,000**:

- **Project Alpha (Short-term):** Inflows: C_1 = INR 70,000, C_2 = INR 60,000.
- **Project Beta (Long-term):** Inflows: C_1 = INR 20,000, C_2 = INR 120,000.

Determine the **Fisher Intersection Rate** r^* where both projects have identical NPVs, and state which project dominates for cost of capital below vs above r^*.

Step-by-Step Analytical Solution:

Step 1 (Formulate Differential Cash Flow Stream ΔC):

At the Fisher rate r^*: $NPV_\alpha(r^*) = NPV_\beta(r^*) \Rightarrow NPV(\Delta C) = 0$.

- $\Delta C_0 = -100{,}000 - (-100{,}000) = 0$.
- $\Delta C_1 = 70{,}000 - 20{,}000 = +50{,}000$.
- $\Delta C_2 = 60{,}000 - 120{,}000 = -60{,}000$.

Step 2 (Solve the Root Equation):

$$50{,}000 / (1 + r^*) - 60{,}000 / (1 + r^*)^2 = 0.$$

Multiplying by $(1 + r^*)^2$:

$$50{,}000(1 + r^*) - 60{,}000 = 0 \Rightarrow 50{,}000(1 + r^*) = 60{,}000$$

$$1 + r^* = 60{,}000 / 50{,}000 = 1.2000.$$

Fisher Intersection Rate: $r^* = 1.2000 - 1 =$ **0.2000 (20.00%)**.

Step 3 (Dominance Decision Matrix):

- For $r <$ **20%** (e.g. 10%): Project Beta dominates (e.g., at 10%: NPV_β = INR 17,355 vs NPV_α = INR 13,223).
- For $r >$ **20%** (e.g. 25%): Project Alpha dominates (e.g., at 25%: $NPV_\alpha = -$INR 5,600 vs $NPV_\beta = -$INR 7,200).
- *Takeaway:* At low discount rates, back-loaded cash flows (Beta) have massive value. As discount rates rise, front-loaded early cash flows (Alpha) suffer far less discount penalty.

9.4 Algorithmic Implementation in Python

Below is the complete implementation from models/capital_budgeting.py, featuring multi-period discounted cash flow summation and sensitivity profiling:

Listing: models/capital_budgeting.py — Net Present Value Engine

```
from typing import List, Tuple

def calculate_npv(cash_flows: List[float], discount_rate: float) -> float:
    """
    Computes Net Present Value: NPV(r) = sum(C_t / (1 + r) ** t).
    """
    if discount_rate <= -1.0:
        raise ValueError("Discount rate must be strictly greater than -100%.")
    return float(sum(cf / ((1.0 + discount_rate) ** t) for t, cf in enumerate(cash_flows)))


def npv_profile(
    cash_flows: List[float], min_rate: float = 0.0, max_rate: float = 0.30, steps: int = 31
) -> List[Tuple[float, float]]:
    """Generates (rate, npv) sensitivity points for profile curve visualization."""
    step_size = (max_rate - min_rate) / (steps - 1)
    profile = []
    for i in range(steps):
        r = min_rate + (i * step_size)
        profile.append((r, calculate_npv(cash_flows, r)))
    return profile
```

9.5 Mathematical Properties Summary

Mathematical Property	Specification & Formula	Analytical Significance
DCF Operator	$\text{NPV}(r) = \sum C_t(1+r)^{-t}$	Linear functional mapping cash streams to present wealth
Rate Monotonicity	$d(\text{NPV})/dr < 0$	Strictly decreasing profile for conventional project structures
Profile Convexity	$d^2(\text{NPV})/dr^2 > 0$	Guarantees uniqueness of roots for standard investment outflows
Reinvestment Axiom	Reinvestment at hurdle r	Avoids unrealistic yield compounding assumptions of IRR
Fisher Intersection	$\text{NPV}_A(r^*) = \text{NPV}_B(r^*)$	Threshold crossover rate for mutually exclusive projects